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Functional Connectivity Associations With Markers of Disease Progression in *GRN* Pathogenic Variant Carriers

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Keywords: frontotemporal dementia with GRN pathogenic variants | functional connectivity | markers of disease progression

ABSTRACT

Objective: Autosomal dominant progranulin (*GRN*) pathogenic variants are a genetic cause of frontotemporal lobar degeneration. Though clinical trials for *GRN*-related therapies are underway, there is an unmet need for biomarkers that can predict symptom onset and track disease progression. We previously showed that asymptomatic *GRN* variant carriers exhibit thalamocortical hyperconnectivity that increases with age, presumably as they are approaching symptom onset. Whether hyperconnectivity arises concomitantly with markers of neurodegeneration remains unclear.

Methods: Utilizing T1 and task-free functional magnetic resonance imaging from 49 asymptomatic and 26 symptomatic *GRN* variant carriers, we determined the relationships between functional connectivity, as measured by voxelwise whole-brain degree, and *GRN*-relevant markers of disease progression, including plasma neurofilament light chain concentrations, cerebrospinal fluid complement C1q and C3b protein levels, obsessive-compulsive disorder symptom severity, and gray matter volume.

Results: Neurofilament light chain concentrations were associated with frontotemporoparietal and thalamic hyperconnectivity in asymptomatic *GRN* variant carriers and extensive regions of atrophy in symptomatic carriers. Complement levels were associated with regions of hyperconnectivity, but not gray matter volume, in symptomatic carriers. Obsessive-compulsive disorder symptom severity was associated with hypoconnectivity across all *GRN* carriers. Asymptomatic carriers with thalamic hyperconnectivity tended to have lower gray matter volume in the bilateral insula and left lateral parietal cortex, early regions of atrophy in *GRN*-frontotemporal dementia.

Interpretation: In asymptomatic carriers, the co-occurrence of hyperconnectivity, high neurofilament light chain, and low gray matter volume suggests that functional hyperconnectivity may portend the onset of clinical decline. These findings point toward hyperconnectivity as an indicator of approaching symptomatic onset.

1 | Introduction

Progranulin (*GRN*) pathogenic variants are a genetic cause of frontotemporal lobar degeneration (FTLD), accounting for 5%–10% of all cases [1, 2]. Though clinical therapy trials are underway for *GRN* variant carriers, there remains an unmet need for biomarkers that predict symptom onset and track disease progression, particularly during asymptomatic and early symptomatic phases. Previous studies in *GRN* variant carriers have explored candidate biomarkers, including structural and task-free fMRI and fluid biomarkers; yet, their cross-sectional relationships during various clinical stages remain unclear.

Despite subtle or absent gray matter volume decreases, asymptomatic genetic frontotemporal dementia (FTD) variant carriers show anomalies in intrinsic connectivity networks as measured by task-free fMRI [3–7]. These studies suggest that functional connectivity measures appear more sensitive than structural imaging metrics during the asymptomatic phase. Across *GRN* studies, asymptomatic carriers have sparse regions of low gray matter volume [3, 8–11], yet exhibit hyperconnectivity in networks whose brain regions degenerate in the most common *GRN* clinical syndromes [5]. In asymptomatic *GRN* (asympGRN), the thalamus exhibits hyperconnectivity in several *GRN* syndrome-associated networks, and hyperconnectivity appears more pronounced in older carriers presumably closer to symptom onset. These findings suggest that hyperconnectivity may signal symptomatic conversion. In contrast, symptomatic *GRN* (sympGRN) exhibits reduced thalamic connectivity [5], yet it remains unclear whether hyperconnectivity anticipates or arises concomitantly with indicators of neurodegeneration.

In addition to the functional connectivity alterations that dynamically characterize the different clinical stages of *GRN*-mediated disease, studies have identified additional markers of disease progression. Specifically, neurofilament light chain (NfL), a protein associated with axonal injury, is elevated in symptomatic carriers compared to asymptomatic carriers and controls, correlates with symptom severity and progression, and identifies asymptomatic carriers at risk for symptomatic phenoconversion [12–17]. Other biomarkers include complement proteins C1q and C3b, part of the innate immunity system, that correlate with worse global cognition in *GRN* variant carriers, and their elevated levels in *Grn* knockout (−/−) mice are associated with thalamocortical hyperexcitability and obsessive grooming behavior. Patients with sporadic and genetic behavioral variant FTD (bvFTD), including *GRN*-bvFTD, may develop obsessive-compulsive behaviors [18]. In parallel, *Grn*−/− mice [19, 20] have shown repetitive grooming behaviors that have been associated with thalamocortical hyperexcitability. Additionally, subtle gray matter volume changes have been detected in asympGRN prior to symptom onset [11, 21]. The relationships between connectivity alterations and these *GRN*-relevant disease progression markers have not been explored.

In the present study, we examined cross-sectional relationships between connectivity measures and *GRN*-relevant disease progression markers in 49 asympGRN and 26 sympGRN as a foundational step toward identifying markers that predict

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symptom onset. Specifically, we determined relationships between voxelwise whole-brain degree connectivity and NfL, complement C1q and C3b, obsessive-compulsive disorder (OCD) symptom severity, and gray matter volume. We hypothesized that measures of disease progression would be associated with functional hyperconnectivity in *GRN* variant carriers.

2 | Methods

2.1 | Participants

Study participants were recruited from the University of California, San Francisco (UCSF) Memory and Aging Center, Hospital Universitario Donostia, and the ARTFL/LEFFTDS Longitudinal Frontotemporal Lobar Degeneration (ALLFTD) study, a multisite genetic frontotemporal dementia study [22]. Clinical diagnoses were rendered at each study site in accordance with standardized criteria.

We identified 75 *GRN* pathogenic variant carriers who tested negative for microtubule-associated protein tau (*MAPT*) variants and the chromosome 9 open reading frame 72 (*C9orf72*) repeat expansion [23] with usable neuroimaging data (see *Image acquisition and preprocessing*; Table 1). The 49 asympGRN were diagnosed as clinically normal and had a global score of 0 on the Clinical Dementia Rating Scale (CDR) plus Behavior and Language domains from the National Alzheimer's Coordinating Center (NACC) FTLD module (CDR plus NACC FTLD) [24, 25]. The 26 sympGRN included the following: 7 with mild cognitive impairment, 1 with a mood disorder and a CDR plus NACC FTLD global score of 0.5, 5 with amnestic dementia of the Alzheimer's type, 9 bvFTD, 1 Parkinson's disease, 1 with a nonspecific primary progressive aphasia (PPA), 1 with nonfluent variant PPA, and 1 with semantic variant PPA. All sympGRN had a CDR plus NACC FTLD global score ≥ 0.5, except for the participant with Parkinson's disease, who scored 0.

We included 71 healthy control participants (HC) for comparison. HC were comprised of noncarrier members of *GRN*, *MAPT*, and *C9orf72* families and unrelated controls who were included as necessary to ensure an age distribution to match the carrier cohorts. All HC were required to have a Mini-Mental State Examination (MMSE) score ≥ 27, and, if unavailable, a corresponding score ≥ 21 on the Montreal Cognitive Assessment (MoCA) [26]. HC had a CDR plus NACC FTLD global score of 0 and no significant white matter lesions or history of neurological disease.

Plasma NfL, cerebrospinal fluid (CSF) complement, and the Yale-Brown Obsessive Compulsive Scale (YBOCS [27]) were available for subsets of asympGRN, sympGRN, and HC. Plasma, CSF, and YBOCS were collected within 180 days of MRI scanning. Due to age differences between carrier groups, separate but overlapping HC groups were matched on age, sex, and handedness to asympGRN and sympGRN for each biomarker and neuroimaging measure association analysis (see Tables S1–S3). While HCs were matched to asympGRN on education,

TABLE 1 | Demographics and clinical measures for *GRN* carriers and healthy controls.

	asympGRN (*n* = 49)	sympGRN (*n* = 26)	HC (*n* = 71)
Age at MRI scan, years	49.7 (12.5)	60.9 (8.1)	52.9 (13.4)
Sex (Male:Female, *n*)	30:19	13:13	39:32
Education, years	17.0 (4.8)	14.0 (3.5)	16.9 (3.0)
Handedness (R:L:A, *n*)	43:6:0	25:1:0	64:7:0
CDR plus NACC FTLD, global score (0–3), median (range)	0 (0–0)	1 (0–2)	0 (0–0)
CDR plus NACC FTLD, sum of boxes (max = 24)	0 (0)	4.8 (4.2)	0 (0)
Montreal Cognitive Assessment	27.1 (2.6)	22.7 (4.1)	27.7 (2.4)
Mini-Mental State Exam	29.3 (1.0)	24.9 (4.4)	29.4 (0.9)
Symptom duration, years	—	2.3 (1.8)	—
Complement, *n*	17	13	24
CSF C1q (ng/mL)	290.8 (130.3)	359.2 (119.2)	347.2 (153.1)
CSF C3b (ng/mL)	3639.7 (1360.9)	4830.8 (1489.0)	4344.1 (1822.3)
Neurofilament light chain, *n*	30	13	48
Plasma neurofilament light chain (pg/mL)	13.8 (13.8)	45.4 (34.0)	13.6 (7.8)
Yale-Brown Obsessive Compulsive Scale (total), *n*	11	9	12
Yale-Brown Obsessive Compulsive Scale (total)	5.2 (4.4)	8.9 (7.1)	6.8 (6.6)

Note: Mean (SD) values are reported for continuous variables, unless otherwise indicated.
Abbreviations: asympGRN = asymptomatic *GRN* pathogenic variant carriers; CDR plus NACC FTLD = Clinical Dementia Rating Scale (CDR) plus Behavior and Language domains from the National Alzheimer's Coordinating Center (NACC) Frontotemporal Lobar Degeneration Module; HC = healthy controls; R:L:A = Right: Left: Ambidextrous; sympGRN = symptomatic *GRN* pathogenic variant carriers.

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sympGRN had fewer years of education compared to matched HCs. For three asympGRN and four HCs, different biomarker data were available at two different visits (e.g., NfL available at one visit, complement available at a different visit), and each visit was included in the corresponding analysis. Throughout both visits, all seven participants of these remained clinically normal. Sample sizes for analyses varied depending on the biomarker, with NfL: asympGRN (30), asympGRN-matched HCs (47), sympGRN (13), sympGRN-matched HCs (33); complement: asympGRN (17), asympGRN-matched HCs (20), sympGRN (13), sympGRN-matched HCs (13); and YBOCS: asympGRN (11), sympGRN (9), HCs (12).

Plasma NfL concentrations were assayed using the Simoa NF-light Advantage Kit from Quanterix on a SiMoA HD-X Analyzer instrument following previous protocols [12, 17, 28] at three processing labs. An ELISA assay was used at one processing lab to determine CSF complement C1q and C3b levels [20].

Institutional Review Boards approved this study; all participants or their surrogates provided informed consent.

2.2 | Image Acquisition and Preprocessing

Each participant underwent a T1-weighted (T1w) structural MRI scan and a T2*-weighted echo-planar imaging tf-fMRI scan. Imaging data were collected on a 3T scanner at each site at the UCSF Memory and Aging Center [4], Fundación Centro de Investigación y Terapias Avanzadas using the Genetic Frontotemporal Dementia Initiative protocol [29], and ALLFTD [22].

2.2.1 | Structural Voxel-Based Morphometry (VBM) Preprocessing

VBM was performed using SPM12 as in previous studies [5, 7]. Briefly, T1-weighted images were preprocessed using standard spatial normalization in the SPM12 segment module, using six standard tissue probability maps with a light cleanup. Standard affine regularization with the International Consortium for Brain Mapping European brain template and warping regularization with the default parameters were used. Images were segmented into gray and white matter images. Gray matter volume (GMV) images were smoothed using an 8 mm full-width at half-maximum (FWHM) isotropic Gaussian kernel.

2.2.2 | Functional MRI (fMRI) Preprocessing

fMRI data were preprocessed using fMRIPrep [30] as described previously [7]. Briefly, the T1w image was corrected for intensity non-uniformity and underwent skull stripping, segmentation, and nonlinear registration to MNI152NLin6Asym (ANTs). The first five fMRI volumes were dropped for magnetic field stabilization. A reference volume and its skull-stripped version were generated using a custom methodology of fMRIPrep and co-registered to the T1w reference using boundary-based registration with six degrees of freedom. Blood-oxygen-level-dependent (BOLD) data were corrected for slice timing, head motion, and susceptibility distortions. Motion-correcting transformations, BOLD-to-T1w transformation, and T1w-to-MNI warp were concatenated and applied in a single step (ANTs; Lanczos interpolation). Data were smoothed with a 6 mm FWHM Gaussian kernel and denoised using non-aggressive ICA-AROMA [31]. Finally, linear detrending, bandpass filtering (0.008–0.08 Hz), and nuisance regression (Friston's 24 motion parameters, mean white matter, and CSF signals) were applied. Participants with inaccurate segmentation or registration, or excessive head motion, defined as (1) mean framewise displacement throughout the acquisition of > 0.5 mm or (2) having less than a total of 4 min of data throughout the entire acquisition without large motion spikes (i.e., > 0.5 mm framewise displacement), were excluded.

2.2.3 | Whole-Brain Degree Analysis

Functional connectivity was assessed using whole-brain degree (WBD) centrality, a graph theoretical metric summarizing each voxel's connectivity to every other voxel in the brain. WBD maps were calculated from preprocessed fMRI images where each voxel contains the sum of partial correlation coefficients (controlling for nuisance parameters) of that voxel to every other voxel [32, 33] using Python version 2.7 with NumPy 1.16.4 and NiBabel 2.2.1 packages [5].

2.2.4 | Harmonization

A customized ComBat approach [7] to remove batch effects of no interest (i.e., each scanner at each site), while preserving biological variability of interest (i.e., age, sex, education, and handedness), was applied to individual participant GMV and WBD maps to harmonize multisite data for subsequent biomarker and neuroimaging measure association analyses. ComBat parameters were estimated in an independent group of 111 HC (Table S4) whose demographic characteristics spanned the age ranges and scanner sites of the study cohorts; they were used to harmonize study cohort data.

2.3 | Statistical Analyses

2.3.1 | Biomarker Levels

Regression analyses compared fluid biomarkers (NfL, C1q, C3b) between asympGRN, sympGRN, and HC, controlling for age and sex. Partial correlations, also controlling for age and sex, assessed biomarker relationships with symptom severity (CDR plus NACC-FTLD sum of boxes (SB) score) across all *GRN* and within sympGRN. NfL was log-transformed, with the processing lab included as a nuisance covariate. Due to limited YBOCS data (asympGRN: $N = 11$, sympGRN: $N = 9$), YBOCS total scores were compared between all *GRN* and HC and correlated with the CDR plus NACC-FTLD-SB score across all *GRN*.

Statistical analyses were completed in R (v 4.2.1). We compared demographic and clinical variables using the chi-squared test, t-test, or Mann–Whitney U test as appropriate. Test statistics were considered significant at $p < 0.05$.

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2.3.2 | Biomarker and Neuroimaging Measure Associations

We examined the relationship between each imaging measure (GMV, WBD) and each biomarker (log(NfL), C1q, C3b, YBOCS-total score, thalamic WBD [GMV only]) using SPM12. For asympGRN and sympGRN separately, voxel-wise correlations identified regions where GMV or WBD significantly correlated with each biomarker. To assess group differences, biomarker by gene status interaction analyses identified voxels where carriers (asympGRN or sympGRN) showed a more positive or negative correlation slope between the biomarker and imaging measure than HC. Relationships between imaging measures and YBOCS scores were examined (1) within all carriers and (2) in the gene status interaction analyses comparing all *GRN* versus HC.

Group-level models were assessed via the general linear model framework in SPM12. Nuisance covariates included age, sex, education, handedness, and total intracranial volume (TIV; GMV only) and eyes-open status during scan acquisition (WBD only). For NfL, the processing lab center was a nuisance covariate. Group-level GMV maps were analyzed using threshold-free cluster enhancement (TFCE [34]) with statistical significance assessed at $p<0.05$, family-wise error (FWE) corrected for multiple comparisons, and a minimum cluster size of $k\geq 10$ voxels, while WBD maps were thresholded with a joint height and extent threshold of $p<0.05$ corrected for multiple comparisons [35].

2.3.3 | Clustering Analysis

Given the likely heterogeneity of thalamic connectivity in asympGRN, we explored whether patterns of thalamic hyper- or hypoconnectivity relate to patterns of gray matter atrophy. To identify subgroups of asympGRN based on thalamic connectivity, we performed K-means clustering on their voxelwise thalamic w-scores.

To calculate w-scores, a voxelwise linear regression was performed on 111 HC (Table S4) WBD maps to determine the standard deviation of the residuals (age, sex, education, handedness, and eyes-open status as nuisance covariates). Each subject's voxelwise w-score was computed as: $w=(-1)$*(subject's value–value predicted for subject age)/sd of residuals of the HC group [36]. A w-score map can be conceptualized as a z-score map with nuisance covariates regressed.

Voxelwise WBD w-scores within the bilateral thalamus were extracted for each asympGRN. K-means clustering was performed using R. The silhouette coefficient determined the optimal number of clusters, with 50 random starts and 100 maximum iterations. For visualization only, principal component analysis was conducted on the same thalamic w-scores, with subgroups defined by K-means clustering plotted along the first two principal components.

To visualize thalamic connectivity patterns, voxelwise mean thalamic w-scores were calculated for each cluster, thresholded at $w\geq 1.96$ and $w\leq -1.96$. Demographic and clinical variables were compared between clusters using the chi-squared test, *t*-test, or Mann–Whitney *U*. To examine GMV differences between clusters, individual GMV w-score maps were created (age, sex, education, handedness, and TIV as nuisance covariates), and GMV w-score frequency maps were generated for each cluster to determine the percentage of participants with abnormally low GMV at baseline ($w\leq -1.65$) in each voxel.

Data are available from the ALLFTD Data management team upon reasonable request (allftd.org/data).

3 | Results

3.1 | NfL Concentrations Were Elevated in the Symptomatic Stage and Associated With Symptom Severity, While Complement Levels Were Similar Across Groups

NfL concentrations were higher in sympGRN (median 45.8 pg/mL [IQR 10.4–69.2]) compared to asympGRN (9.70 pg/mL [7.35–14.4], $p=0.003$) and HC (11.7 pg/mL [8.47–15.6], $p<0.001$, Figure 1A). However, NfL concentrations did not differ significantly between asympGRN and HC ($p>0.05$). Yet, across all *GRN* carriers and within sympGRN, higher NfL concentrations were associated with increasing symptom severity as measured by the CDR plus NACC-FTLD box score (across *GRN*: $\beta=0.12$, 95% CI 0.07–0.17, $p<0.001$; sympGRN: $\beta=0.13$, 95% CI 0.04–0.23, $p=0.01$).

C1q and C3b concentrations did not differ between asympGRN (C1q: 279 ng/mL [235–354]; C3b: 3914 ng/mL [2812–4510]), sympGRN (C1q: 334 ng/mL [274–439]; C3B: 5250 ng/mL [3913–5812]), and HC (C1q: 322 ng/mL [213–453]; C3B: 3873 ng/mL [3223–4439], $p=0.30$ for C1q and $p=0.10$ for C3b, Figure 1B,C) after controlling for age and sex. Further, C1q and C3b concentrations did not associate with symptom severity (CDR plus NACC-FTLD box score) across all carriers (C1q: $\beta=0.28$, 95% CI −13.93 to 14.47, $p=0.97$; C3b: $\beta=145.01$, 95% CI −11.73 to 301.75, $p=0.07$) or within sympGRN (C1q: $\beta=-8.71$, 95% CI −27.69 to 10.28, $p=0.33$; C3b: $\beta=57.04$, 95% CI −134.72 to 248.80, $p=0.52$).

3.2 | Higher NfL Concentrations Were Associated With Regions of Higher Connectivity in asympGRN and With Gray Matter Atrophy in sympGRN

Higher NfL concentrations were associated with higher WBD, a metric summarizing each voxel's connectivity to the remaining voxels, in medial frontal regions and the brainstem in asympGRN (Figure 2A). To determine whether the relationship between NfL concentrations and WBD differed in asympGRN compared to HC, NfL by gene status interaction analyses were performed. Compared to HC, asympGRN exhibited a more positive relationship between NfL concentrations and WBD in widespread regions, including the bilateral medial frontal cortex, anterior insula, and thalamus (Figure 2B). In asympGRN, NfL concentrations showed no correlation or gene status interaction with GMV.

In contrast to asympGRN, NfL concentrations in sympGRN did not show a correlation or gene status interaction with WBD and were also not correlated with GMV. However, a gene status interaction was found for GMV, with sympGRN exhibiting a more

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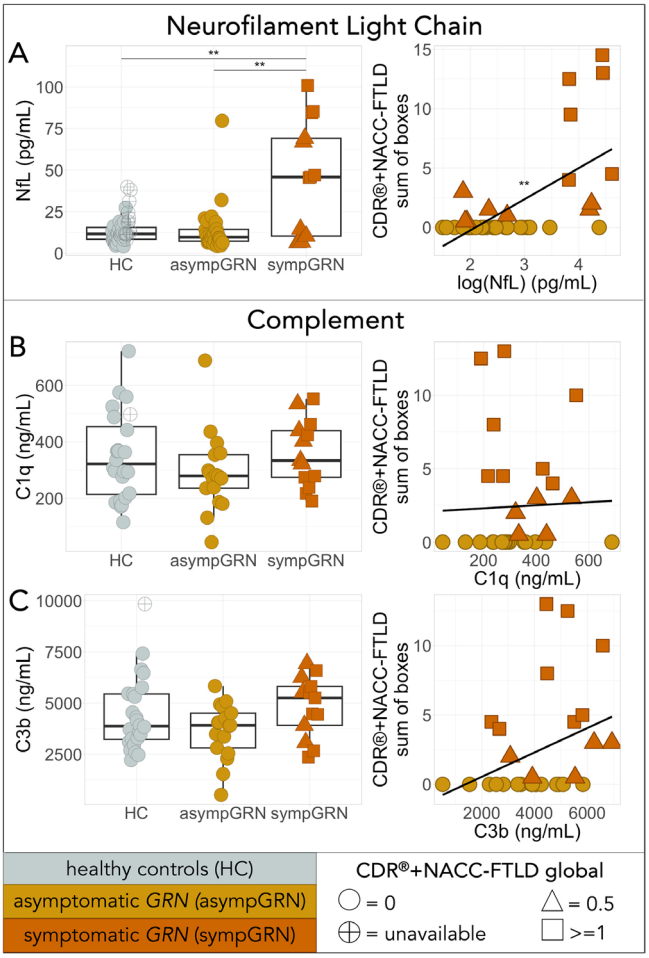

FIGURE 1 | NfL concentrations were elevated in the symptomatic stage and associated with greater symptom severity, while complement levels were similar across groups. (A) NfL concentrations were higher in sympGRN ($n=13$; orange) compared to asympGRN ($n=30$; yellow) and HC ($n=48$; gray). Higher NfL was associated with higher CDR plus NACC-FTLD box scores across all carriers ($\beta=0.12$, 95% CI 0.07–0.17, $p<0.001$). (B) C1q levels did not differ between sympGRN ($n=13$), asympGRN ($n=17$), and HC ($n=24$) and were not associated with CDR plus NACC-FTLD box scores ($\beta=0.28$, 95% CI −13.93 to 14.47, $p=0.97$). (C) C3b levels did not differ between sympGRN ($n=13$), asympGRN ($n=17$), and HC ($n=24$) and were not associated with CDR plus NACC-FTLD box scores ($\beta=145.01$, 95% CI −11.73 to 301.75, $p=0.07$). asympGRN = asymptomatic *GRN* pathogenic variant carriers; CDR plus NACC FTLD = Clinical Dementia Rating Scale (CDR) plus Behavior and Language domains from the National Alzheimer's Coordinating Center (NACC) Frontotemporal Lobar Degeneration Module; HC = healthy controls; NfL = neurofilament light chain; sympGRN = symptomatic *GRN* pathogenic variant carriers.

negative relationship between NfL concentrations and GMV in widespread regions compared to HC (Figure 2C).

3.3 | Complement Levels Were Associated With WBD but Not Gray Matter

In asympGRN, higher C1q levels were associated with lower WBD in medial frontal and cerebellar regions, while also showing a more positive relationship with WBD in the right lateral frontal cortex, including the right frontal pole and inferior frontal gyrus compared to HC (Figure 3A). However, C3b did not show a correlation or gene status interaction with WBD. Additionally, neither C1q nor C3b showed correlations or gene status interactions with GMV in asympGRN.

While no associations were observed between C1q and WBD in sympGRN, this group exhibited a more positive relationship between C1q and WBD in the left lateral and medial frontal cortex and precuneus compared to HC (Figure 3B). C3b showed both a correlation and gene status interaction with WBD in sympGRN. Specifically, higher C3b was associated with higher WBD in widespread regions, including bilateral dorsolateral/medial frontal and insular cortices and thalami (Figure 3C). Similarly, sympGRN had a more positive relationship between C3b levels and WBD in the left dorsolateral frontal, left lateral parietal, and bilateral anterior thalami compared to HC (Figure 3C). Similar to asympGRN, neither C1q nor C3b showed correlations or gene status interactions with GMV in sympGRN.

3.4 | Higher OCD Symptoms Associated With Regions of Lower Functional Connectivity Across All *GRN*

Given the relatively few numbers of participants with YBOCS scores, data were analyzed across all *GRN*. Carriers did not differ from HC in YBOCS total score (median 5.5 [IQR 1.75–11.5] vs. 6.5 [0.75–9], $p=0.97$). YBOCS total was not correlated with symptom severity (CDR plus NACC-FTLD-SB; $\rho=0.09$, 95% CI −0.38 to 0.53, $p=0.70$).

Greater OCD symptoms (YBOCS) were associated with lower WBD in bilateral anterior and posterior cingulate, insula, middle temporal gyrus, precuneus, and caudate (Figure 4A). Also, carriers had a more negative relationship between OCD symptoms and WBD compared to HC in bilateral middle temporal gyrus, precuneus, and occipital cortex, and right precentral gyrus, hippocampus, and putamen (Figure 4B). OCD symptom severity did not show a significant correlation or gene status interaction with GMV in *GRN* carriers.

3.5 | AsympGRN With Thalamic Hyperconnectivity Have Regions of Lower Gray Matter Volume

Mean thalamic WBD showed no correlation or gene status interaction with voxelwise GMV in asympGRN or sympGRN. Since asympGRN are heterogeneous with respect to symptomatic onset, we reasoned that a subset may have thalamic hyperconnectivity. Thus, we explored how different levels of thalamic connectivity may be associated with different gray matter atrophy patterns.

K-means clustering on 49 asympGRN's voxelwise thalamic w-scores identified two clusters based on thalamic connectivity: cluster 1 included 41 participants and cluster 2 included 8 participants (Figure 5A). AsympGRN in cluster 2 were older with fewer years of education than cluster 1, but did not differ in sex, handedness, CDR plus NACC-FTLD global or sum of boxes, MoCA score, or NfL concentrations (Figure 5B). Although

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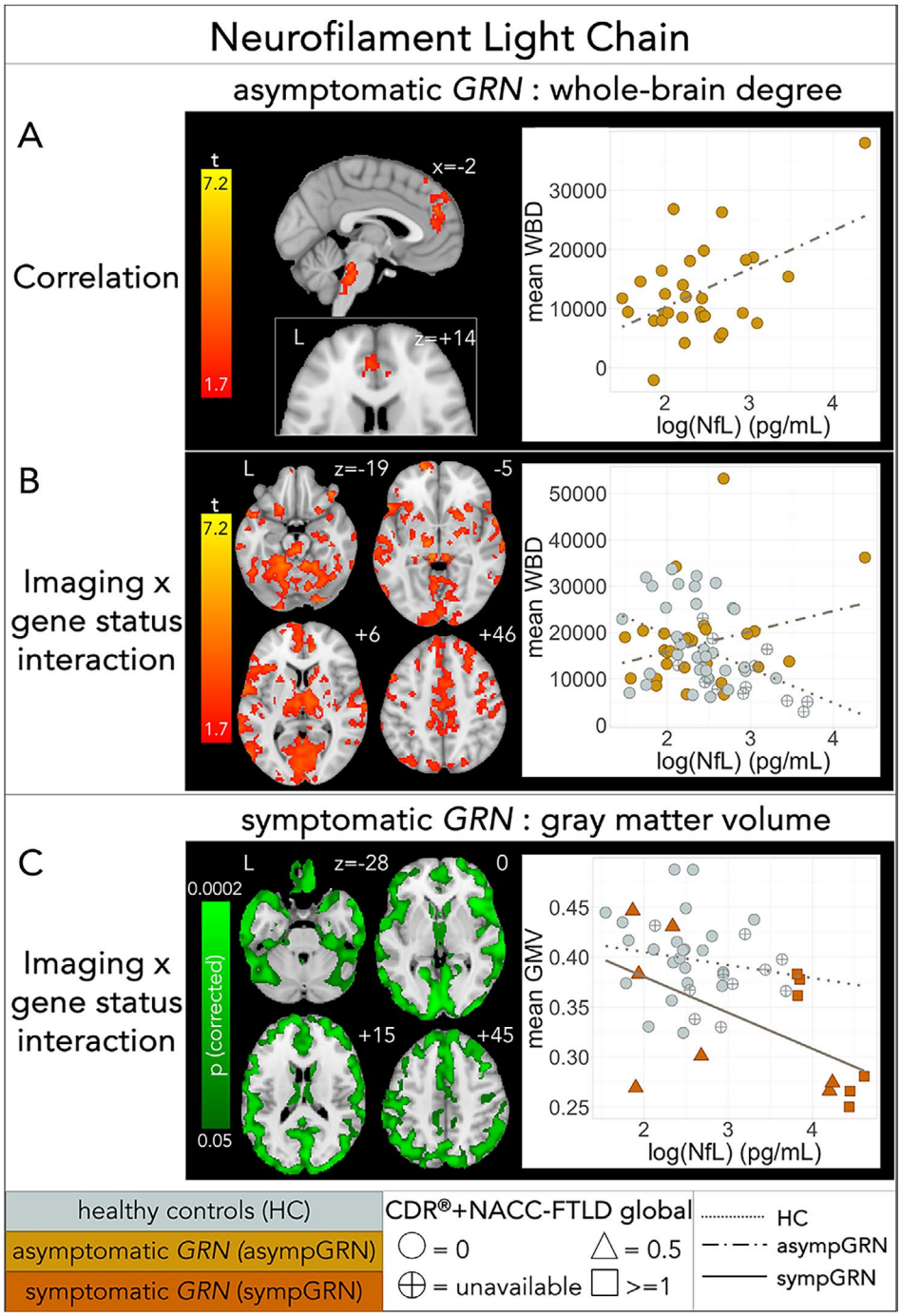

FIGURE 2 | Higher NfL concentrations were associated with regions of higher functional connectivity in asympGRN and with gray matter atrophy in sympGRN. (A) Higher NfL concentrations were associated with higher WBD in medial frontal regions and the brainstem in asympGRN ($n = 30$). (B) AsympGRN exhibited a more positive relationship between NfL concentrations and WBD in widespread regions, including the bilateral medial frontal cortex, anterior insula, and thalamus, compared to HC ($n = 47$). (C) SympGRN ($n = 13$) exhibited a more negative relationship between NfL concentrations and GMV in widespread regions compared to HC ($n = 13$). WBD results are thresholded at a joint height and extent threshold of $p < 0.05$; GMV results are thresholded using threshold-free cluster enhancement (TFCE) with $p < 0.05$, FWE-corrected, and a minimum cluster size of $k \geq 10$ voxels. All results are superimposed on the Montreal Neurological Institute brain template. The mean WBD or GMV values represent the average WBD or GMV within significant voxels. Scatterplots are provided for visualization purposes only and should not be used for statistical inference. asympGRN = asymptomatic *GRN* pathogenic variant carriers; CDR plus NACC FTLD = Clinical Dementia Rating Scale (CDR) plus Behavior and Language domains from the National Alzheimer's Coordinating Center (NACC) Frontotemporal Lobar Degeneration Module; GMV = gray matter volume; HC = healthy controls; NfL = neurofilament light chain; sympGRN = symptomatic *GRN* pathogenic variant carriers; WBD = whole-brain degree.

cluster 2 had lower MMSE scores (median 28 [IQR 28–29]) compared to cluster 1 (30 [29–30], $p = 0.04$), these mean scores are considered clinically normal (≥ 27). Mean thalamic WBD connectivity w-score maps indicated that cluster 1 exhibited no voxels of thalamic hyper- or hypoconnectivity, while cluster 2 demonstrated thalamic hyperconnectivity (Figure 5C).

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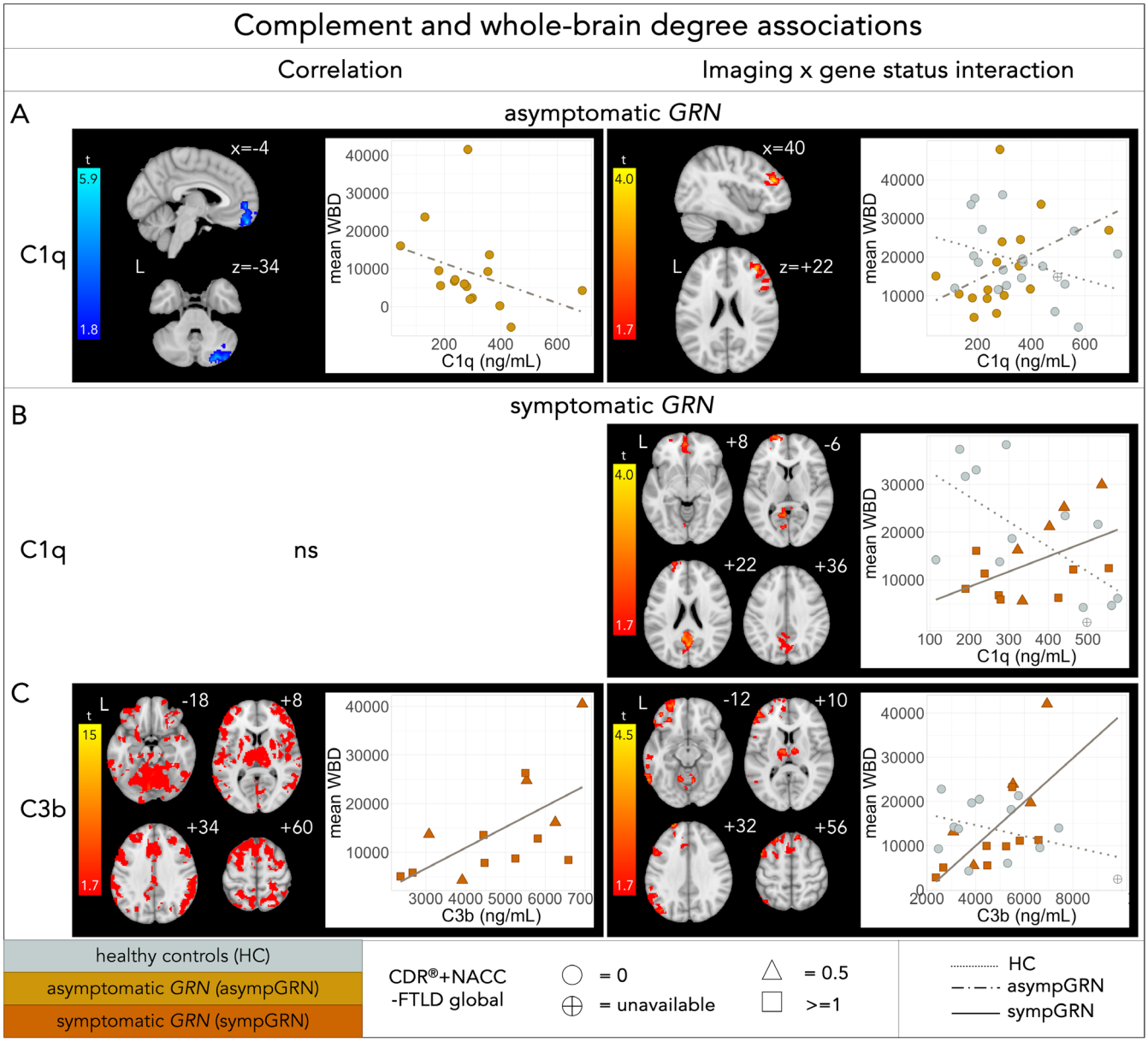

FIGURE 3 | Complement concentrations were associated with WBD but not GMV loss in asympGRN and sympGRN. (A) In asympGRN ($n = 17$), higher C1q levels were associated with lower WBD in medial frontal and cerebellar regions, while also showing a more positive relationship with WBD in the right lateral frontal cortex compared to HC ($n = 20$). (B) No significant associations were observed between C1q and WBD in sympGRN ($n = 13$). However, this group exhibited a significantly more positive relationship between C1q and WBD in the left lateral and medial frontal cortex and precuneus compared to HC ($n = 13$). (C) In sympGRN ($n = 13$), higher C3b was associated with higher WBD in widespread cortical and subcortical regions, including bilateral dorsolateral/medial frontal and insular cortices and thalami. SympGRN also had a more positive relationship between C3b concentrations and WBD in the left dorsolateral frontal, left lateral parietal, and bilateral anterior thalami compared to HC ($n = 13$). WBD results are thresholded at a joint height and extent threshold of $p < 0.05$ and superimposed on the Montreal Neurological Institute brain template. The mean WBD values represent the average WBD within significant voxels. Scatterplots are provided for visualization purposes only and should not be used for statistical inference. asympGRN = asymptomatic *GRN* pathogenic variant carriers; CDR plus NACC FTLD = Clinical Dementia Rating Scale (CDR) plus Behavior and Language domains from the National Alzheimer's Coordinating Center (NACC) Frontotemporal Lobar Degeneration Module; GMV = gray matter volume; HC = healthy controls; ns = not significant; sympGRN = symptomatic *GRN* pathogenic variant carriers; WBD = whole-brain degree.

To better understand thalamic connectivity and GMV relationships, we looked at the percentage of subjects showing low GMV ($w \leq -1.65$) within each cluster (Figure 5D). Notably, while up to 34.2% of asympGRN in cluster 1 exhibited low GMV in certain voxels, the proportion rose to 75% among cluster 2. Specifically, 34.2% of cluster 1 asympGRN had low GMV in voxels in the left middle temporal gyrus, and between 24.3% and 29.3% had low GMV in the bilateral middle frontal gyrus, left inferior temporal gyrus, right temporal pole, left medio-dorsal thalamus, and bilateral cerebellum. At this same threshold, 75% of cluster 2 asympGRN had low GMV in voxels in the left lateral parietal cortex and bilateral cerebellum, 62.5% had low GMV in the left

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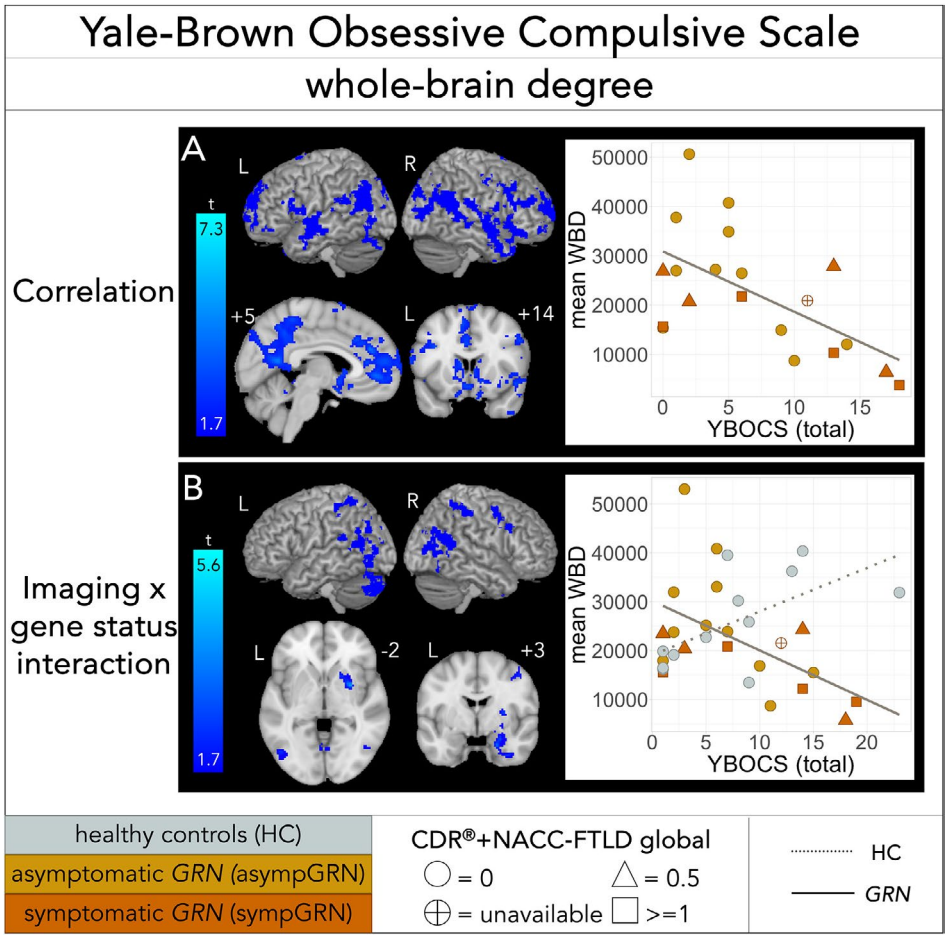

FIGURE 4 | Higher OCD symptoms associated with regions of lower GMV and lower WBD across all *GRN*. (A) A greater degree of OCD symptoms as measured by the Yale-Brown Obsessive Compulsive Scale total score was associated with lower WBD in widespread regions including the bilateral anterior and posterior cingulate, insula, middle temporal gyrus, precuneus, and caudate. (B) *GRN* variant carriers ($n = 20$) had a more negative relationship between OCD symptoms and WBD compared to HCs ($n = 12$) in regions including bilateral middle temporal gyrus, precuneus and occipital cortex, and right precentral gyrus, hippocampus, and putamen. WBD results are thresholded at a joint height and extent threshold of $p < 0.05$ and superimposed on the Montreal Neurological Institute brain template. The mean WBD values represent the average WBD within significant voxels. Scatterplots are provided for visualization purposes only and should not be used for statistical inference. asympGRN = asymptomatic *GRN* pathogenic variant carriers; CDR plus NACC FTLD = Clinical Dementia Rating Scale (CDR) plus Behavior and Language domains from the National Alzheimer's Coordinating Center (NACC) Frontotemporal Lobar Degeneration Module; GMV = gray matter volume; HC = healthy controls; OCD = obsessive compulsive disorder; sympGRN = symptomatic *GRN* pathogenic variant carriers; WBD = whole-brain degree; YBOCS = Yale-Brown Obsessive Compulsive Scale.

anterior insula, right lateral frontal cortex, right middle temporal gyrus, and left temporal pole, and 50% had low GMV in the right anterior and posterior insula, right inferior frontal gyrus, and right temporal pole.

4 | Discussion

The present study identified associations between higher functional connectivity and indicators of disease progression in both asymptomatic and symptomatic phases of *GRN*-FTD. Higher NfL concentrations were linked with higher WBD in asympGRN, while in sympGRN, higher connectivity was associated with higher levels of complement C1q and C3b, known to cause abnormal synaptic pruning and thalamocortical hyperexcitability in *Grn*−/− mice [20]. Obsessive-compulsive symptom severity in carriers was linked to widespread reduced functional connectivity. The asympGRN with higher thalamic functional connectivity also tended to have more GMV loss in parietal and insular cortices, regions that deteriorate in FTD. Overall, these findings suggest that in asympGRN, hyperconnectivity may portend the onset of the symptomatic phase.

4.1 | NfL Concentrations Were Associated With Higher Connectivity in asympGRN, but Not sympGRN

The association between higher NfL concentrations and regions of higher functional connectivity in asympGRN suggests that elevated connectivity may be a sign of early-stage disease. NfL concentrations are linked to greater symptom severity and predict increased likelihood of future clinical decline in asymptomatic FTD variant carriers [16]. The observed relationship between

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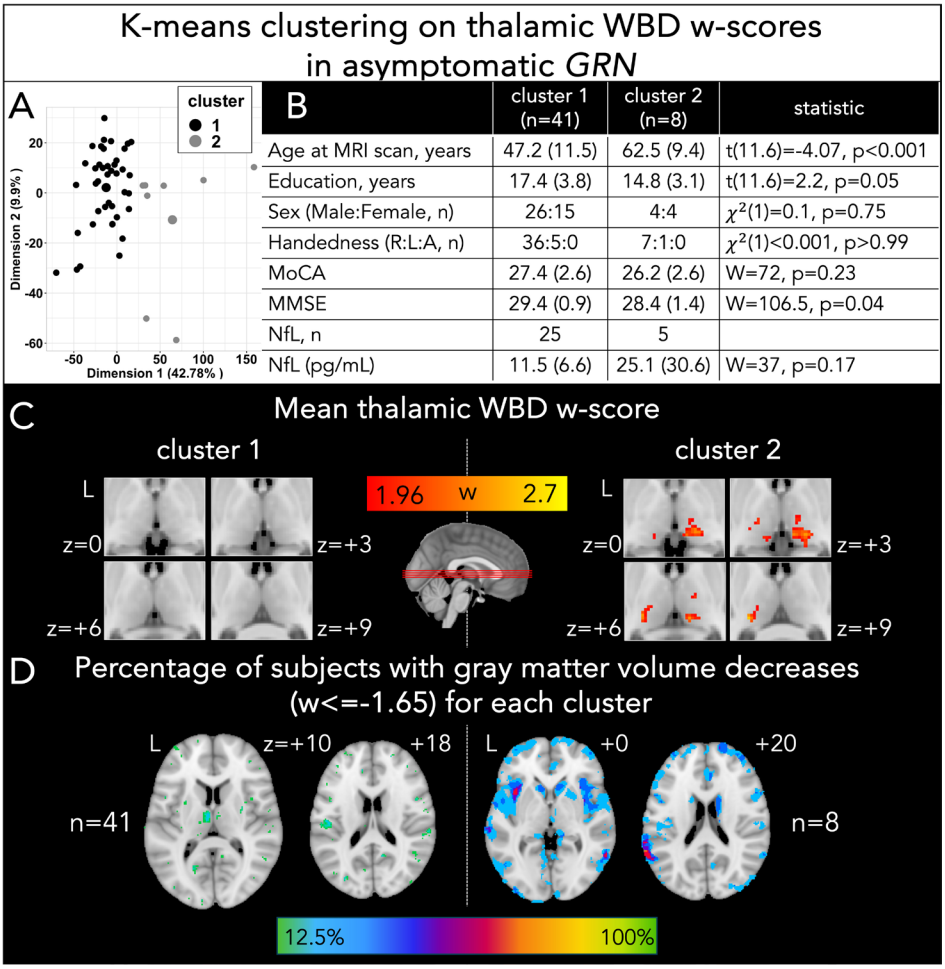

B	cluster 1 (n=41)	cluster 2 (n=8)	statistic
Age at MRI scan, years	47.2 (11.5)	62.5 (9.4)	t(11.6)=-4.07, p<0.001
Education, years	17.4 (3.8)	14.8 (3.1)	t(11.6)=2.2, p=0.05
Sex (Male:Female, n)	26:15	4:4	$\chi^2(1)$=0.1, p=0.75
Handedness (R:L:A, n)	36:5:0	7:1:0	$\chi^2(1)$<0.001, p>0.99
MoCA	27.4 (2.6)	26.2 (2.6)	W=72, p=0.23
MMSE	29.4 (0.9)	28.4 (1.4)	W=106.5, p=0.04
NfL, n	25	5	
NfL (pg/mL)	11.5 (6.6)	25.1 (30.6)	W=37, p=0.17

FIGURE 5 | Clustering analysis based on voxel-wise thalamic functional connectivity identifies asympGRN with thalamic hyperconnectivity who also have regions of lower GMV. (A) K-means clustering based on voxel-wise thalamic whole-brain degree w-scores yields two asympGRN subgroups: cluster 1, which included 41 subjects, and cluster 2, which included 8 subjects. For post hoc visualization only, a principal component analysis based on the same voxelwise thalamic WBD values is conducted, and the 2 subgroups defined by *K*-means clustering are plotted along the 2 principal components that explain the most variance (principal component 1 vs. principal component 2). (B) Demographic table comparing clusters shows that participants in cluster 2 were significantly older and had fewer years of education than cluster 1, but the two clusters did not differ in sex, handedness, CDR plus NACC-FTLD global or sum of boxes, MoCA score, or NfL concentrations. (C) Mean w-score thalamic WBD connectivity maps for each cluster thresholded at $w \geq 1.96$ and $w \leq -1.96$ indicate that cluster 1 exhibited no voxels of hyper- or hypoconnectivity within the thalamus, while cluster 2 demonstrated hyperconnectivity within the thalamus. (D) The percentage of subjects with GMV decreases ($w \leq 1.65$) for each cluster thresholded at 12.5%, the threshold at which at least two participants (based on cluster 2) had low GMV within a voxel. Up to 34.2% of asympGRN in cluster 1 had overlapping voxels of low GMV, while up to 75% of asympGRN in cluster 2 had overlapping voxels of low GMV. asympGRN = asymptomatic *GRN* pathogenic variant carriers; CDR plus NACC FTLD = Clinical Dementia Rating Scale (CDR) plus Behavior and Language domains from the National Alzheimer's Coordinating Center (NACC) Frontotemporal Lobar Degeneration Module; HC = healthy controls; MMSE = Mini-Mental State Examination; MoCA = Montreal Cognitive Assessment; NfL = neurofilament light chain; R:L:A = Right: Left: Ambidextrous; WBD = whole-brain degree.

NfL and functional connectivity in asympGRN (Figure 2A,B) aligns with our previous research demonstrating that connectivity increases with age in asympGRN, thus appearing more prominent in those who may be closer to symptom onset [5]. Notably, the association between NfL and functional connectivity, rather than GMV, underscores the sensitivity of functional connectivity measures over structural measures in detecting alterations in asymptomatic carriers [3–7].

Compared to controls, sympGRN had elevated NfL concentrations and a stronger association of higher NfL with widespread gray matter atrophy (Figure 2C). This finding is consistent with studies showing that NfL is abnormally elevated and associated with atrophy in genetic FTD, including *GRN*-FTD [13–17, 37]. When examining sympGRN alone, we did not identify a correlation between NfL and GMV. Due to the relatively small size of this group, we may have lacked power to detect a relationship. Upon comparing sympGRN with HC, a significant relationship was observed, and including HC in the interaction model may have enhanced statistical power. In contrast to asympGRN, sympGRN lacked associations between NfL and functional connectivity. The correlation between NfL and atrophy suggests that for sympGRN, assessing GMV may be more optimal than functional connectivity, aligning with the finding that structural

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MRI measures excel in discriminating sympGRN from controls while functional hyperconnectivity assessment demonstrates superiority over structural measures for asympGRN [6].

4.2 | Complement Levels Were Associated With Higher Functional Connectivity in sympGRN and, to a Lesser Extent, asympGRN

In sympGRN, higher C1q and C3b were associated with increased connectivity in cortical and subcortical regions, including the thalamus (Figure 3B,C). This finding is consistent with a study of *Grn*−/− mice showing that progranulin loss causes excessive complement activation, leading to enhanced microglial pruning of thalamic inhibitory neurons, resulting in thalamocortical hyperexcitability [20]. Thus, this complement-associated hyperconnectivity in sympGRN offers a compelling link between complement dysregulation and aberrant functional connectivity in sympGRN.

The relationship between complement proteins and connectivity was more complicated in asympGRN. C1q was associated with both lower connectivity (correlation) and higher connectivity (gene status interaction), while C3b showed no significant relationships within asympGRN (Figure 3A). A study of seven asymptomatic carriers (four *GRN* and three *C9orf72* repeat expansion) who were diagnosed with FTD at follow-up found no relationship between baseline complement proteins and time to symptomatic onset [38]. This suggests that C1q and C3b are not optimal for detecting changes in the asymptomatic stage and may explain the lack of a consistent relationship between C1q and C3b in asympGRN. In contrast to previous studies [20, 38], we did not identify a correlation between C1q or C3b levels and symptom severity (CDR plus NACC-FTLD-SB) or differences in C1q or C3b levels between sympGRN, asympGRN, and HC. This aligns with a study demonstrating that complement levels remain consistent across disease stages in *GRN* [38].

4.3 | OCD Symptoms Are Associated With Regions of Lower Connectivity Across All Carriers

We found a greater degree of OCD symptoms associated with lower anterior cingulate and right lateral occipital volume and frontotemporostriatal hypoconnectivity (Figure 4), regions that overlap with OCD-associated atrophy in FTD. OCD symptoms in FTD have been associated with striatal and lateral temporal atrophy [39], as well as anterior cingulate, bilateral amygdala, and hippocampal atrophy [40]. Notably, OCD behaviors in patients with bvFTD have been linked to increased default mode network connectivity [41]. Specific networks may exhibit distinct relationships with OCD behaviors, potentially differing from the whole-brain degree measure employed in the current study, which reflects the overall functional connectivity of a region.

In contrast to the observed hypoconnectivity associated with OCD symptoms in the present study, *Grn*−/− mice exhibit thalamocortical hyperexcitability [20]. Rodents with genetic FTD variants show repetitive behaviors, and in *Grn*−/− mice, medium spiny neuron hyperexcitability in the nucleus accumbens is linked to obsessive grooming [19]. It is worth noting that fMRI hyperconnectivity measures in human studies measure different phenomena from the hyperexcitability observed in *Grn*−/− mice. fMRI hyperconnectivity reflects increased temporal coherence of BOLD signal between two brain regions in a clinical cohort versus controls [42]. Hyperexcitability in thalamic brain slices from *Grn*−/− mice was defined as more numerous action potentials recorded extracellularly compared to *Grn*+/+ mice [20]. Another possibility is that the disease stages of our human and the mouse cohorts may be discrepant, influencing the results.

Though our overall finding was OCD-associated hypoconnectivity across all carriers, this may be driven by the subset of sympGRN who develop OCD symptoms. We, however, did not examine OCD-imaging associations in sympGRN alone due to limited sample size. While hyperconnectivity was a feature of the asymptomatic stage in our present and previous work, sympGRN develops thalamic hypoconnectivity [5]. Neurodegenerative diseases such as behavioral variant FTD and Alzheimer's disease feature reduced connectivity during the symptomatic stage [43–46], commonly revealing a pattern of hypoconnectivity within atrophic regions. For *GRN*, larger datasets would clarify whether OCD symptom severity shows different relationships with connectivity in carriers experiencing different clinical stages (e.g., hyperconnectivity for asympGRN and hypoconnectivity for sympGRN).

4.4 | AsympGRN With Thalamic Hyperconnectivity Have Low Gray Matter in FTD-Associated Regions

In asympGRN, increased thalamic connectivity may herald the approach toward symptom onset. Using cluster analysis, we identified two asympGRN subgroups based on thalamic connectivity (Figure 5). Notably, one asymptomatic subgroup had thalamic hyperconnectivity, while the other lacked thalamic connectivity abnormalities. Importantly, the asympGRN subgroup with thalamic hyperconnectivity was older, with 75% showing regions of low left lateral parietal volume and 62.5% with low GMV in the left anterior insula, right lateral frontal cortex, and left temporal pole, key regions that deteriorate in *GRN*-FTD. In contrast, only 34.2% of the asympGRN subgroup lacking thalamic connectivity abnormalities had low GMV in overlapping regions. These findings align with a prior observation that functional connectivity, including thalamic, increases in older asympGRN, who are likely closer to symptom onset [5]. Consequently, higher connectivity may signal conversion to the symptomatic phase. Studies reveal that hyperconnectivity relates to increased risk of conversion and is a precursor to the symptomatic stage [47–50]. The present results reveal that assessing hyperconnectivity within critical FTD regions may identify asympGRN at risk for symptom conversion, which would aid cohort selection for clinical trials.

5 | Conclusions

We discovered that functional connectivity alterations are associated with markers of disease progression throughout clinical

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stages of *GRN*-FTD, providing a foundational step toward identifying biomarkers that predict symptom onset. In asympGRN, the co-occurrence of hyperconnectivity, high NfL, and low gray matter suggests that hyperconnectivity may portend the onset of the symptomatic phase. Future longitudinal studies are needed to validate functional hyperconnectivity as a biomarker in asymptomatic carriers. In sympGRN, hyperconnectivity was linked to higher levels of complement proteins. Thus, each marker appears poised to detect dynamic changes during different phases.

Our main limitation was that the actual symptom onset of most asympGRN was unknown. Future studies will confirm the extent to which individuals exhibiting higher connectivity are at risk of conversion to the symptomatic phase. Although education was included as a covariate in our analyses, the sympGRN group in our study had fewer years of education compared to HC, which may have influenced the results. Additionally, the sympGRN sample size in the NfL and complement analyses was relatively limited, which may limit the robustness and generalizability of these findings. Future studies with larger sample sizes are needed to replicate these observations and assess the reliability of both neuroimaging and fluid markers once sufficient longitudinal data become available.

In sum, our findings suggest that assessing functional connectivity alterations in *GRN* has the potential to identify carriers who appear at risk for progression to the symptomatic stage. Finding biomarkers that detect early changes and track disease progression is critical for the success of future clinical trials.

Author Contributions

T.M.F. and S.E.L. contributed to the conception and design of the study. The ARTFL/LEFFTDS/ALLFTD Consortia, M.L.M, M.L.G.-T., J.S.Y, and B.L.M. contributed to data acquisition; T.M.F., S.A.C, S.H., L.Z., J.A.B., A.J.L., W.W.S., and S.E.L. contributed to the analysis of data. All authors contributed to manuscript writing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from ALLFTD. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the ALLFTD Data management team upon reasonable request (allftd.org/data).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Demographics and clinical measures for *GRN* variant carriers and healthy controls included in NfL analyses. **Table S2:** Demographics and clinical measures for *GRN* variant carriers and healthy controls included in complement analyses. **Table S3:** Demographics and clinical measures for *GRN* variant carriers and healthy controls included in Yale-Brown Obsessive Compulsive Scale (YBOCS) analyses. **Table S4:** Demographics, clinical measures, and site counts for healthy controls used for ComBat harmonization and w-regression models, and site counts for the study cohort.

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